

http-api-delta

HTTP API Contract — Delta for Knowledge Areas & Deep Linking

Feature: specs/002-knowledge-areas-deep-linking | **Date:** 2026-09-01

This is a **delta**. The platform's existing interface contract remains authoritative for everything it already covers — the response envelope, the three-state contract, the header set, the error taxonomy, the malformed-request rules. This document specifies only what this feature **adds** or **changes**, and it restates none of the existing contract, so the two cannot drift.

1. Binding rules inherited, not restated

Every endpoint below inherits, without exception:

1. The **shared envelope** and its three states — a determined positive, a determined negative, and could-not-determine. A degraded backend is never rendered as an empty result set.
2. The **four resolution outcomes** — found, redacted, not present, could not determine — mapped to their existing status codes. Redacted is not absent, and could-not-determine is not absent.
3. **Malformed requests are a fourth thing**, not a fourth state, and use the existing error shape rather than borrowing the three-state vocabulary.
4. The **generation header**, so a client can tell which index generation answered it.
5. **No caching** on the three-state endpoints.

2. The route-manifest obligation — read this before adding any endpoint

The platform's server-unity verifier reads a route manifest and a contract, and it fails when a contracted endpoint has no manifest row. Its counters are PASS, FAIL, UNDETERMINED and DEBT, and its exit is three-valued: an undetermined result is **not** a pass.

Therefore, for every endpoint in §3:

- **R1** — it gets a row in the route manifest **and** a section here. An endpoint built but undeclared fails the verifier by construction. This is deliberate and must not be “fixed” by loosening the verifier.
- **R2** — an endpoint declared here but not yet built answers as a **not-built route**: 404, with **no** status field, **no** error field, and **no** single-page-application shell in the body. The absence of a body is how a client distinguishes “*this route does not exist*” from “*the thing you asked for does not exist*”. Such a row is counted as **DEBT**, which is a standing visible notice rather than a failure — a known gap must never go quietly green.
- **R3** — an endpoint that returns the SPA shell instead of its own answer is a **failure**, not a cosmetic issue. The existing verifier already checks for this because it has already happened.

And separately, for every endpoint in §4 — the routes this feature **changed** rather than added:

- **R1b** — its manifest row's contract citation names **both** documents: the contracts/http-api.md section the row already cited **and** the §4 subsection here that changed it, written as a compound token (3.7+002.4.1, 3.6+002.4.2, 3.11+002.4.4). A row naming only the older section **still looks complete to a reviewer**, while the behaviour underneath it has since changed — and R1 cannot see that, by construction: R1 asks whether a row exists, and every one of these rows existed throughout this feature. The failure being guarded is therefore not an **absent** row but a **stale** one, and staleness and absence are not the same defect.

The compound form is required rather than a replacement, and both halves are load-bearing. Dropping the 001 section strands the server-unity verifier, which walks contracts/http-api.md §3 and prints that column. Adding a *second* row for the same method and path would leave one route described in two places that are free to disagree — duplication is not documentation. One row, both citations.

R1b IS satisfied today — 4 of 4. The paragraph that stood here is superseded and is kept, struck through, because how the gap was found matters more than the gap. ~~§4 contains four changed endpoints and only three carry a compound token; measured 2026-09-03, the /api/ask rows cite a bare 3.10, with nothing naming §4.3. WITHDRAWN, same day.~~ The answering row now reads 3.10+002.4.3 alongside 3.6+002.4.2, 3.7+002.4.1 and 3.11+002.4.4; verify-server-unity.sh was re-run after the edit at PASS=35 FAIL=0 UNDET=0 DEBT=4, exit 0.

The gap was invisible to the check that existed to find it, and that is the durable lesson. The gate asserting R1b is G-KG-1-~~changed~~, and the published closure check could not see its identifier at all: the extractor G-[A-Z]+-[0-9]+ stops at the digits, so G-KG-1-~~changed~~ was read as G-KG-1 — a different gate, already attached — and the check reported unattached: 0 while this violation stood. Corrected to G-[A-Z]+-[0-9]+(-[a-z]+)* with the boundary guard ([^0-9A-Za-z-]|\$), the check sees 19 identifiers where it saw 18.

A claim written here earlier the same day — that G-KG-1-~~changed~~ “remains unbuilt” — is WITHDRAWN. The gate exists, is registered, is mutation-proved, and PASSES. Measured 2026-09-03: it enumerates 002.4.1, 002.4.2 and 002.4.4, and **002.4.3 appears nowhere in it or in its prover.** §4 has four changed endpoints; the gate asserts over three.

It was green throughout, for the endpoint it does not look at. That is the finding worth carrying: not a broken instrument, but a correct one whose **hand-listed coverage set** was short by exactly the row that was wrong. A gate’s want table is itself an assertion, and nothing was checking that assertion. **Prefer a coverage set DERIVED from §4’s own subsection headings over a hand-maintained list** — a derived set cannot silently omit a section that exists. Until then the outstanding work is one row plus a fourth mutation case, owned by **T143**, which stays unticked.

That citation was stale rather than merely terse: §4.3’s C4.3.3 and C4.3.4 have since been built, and the route emits two response members it did not emit under specs/001 alone — the decline `does_not_answer` and the `could-not-determine` code `question_verification_unavailable`, both now declared in `contracts/http-api.md` §5.5 and §5.3. R1b records this as an **open obligation on /api/ask**; it does not pardon it, and it must not be closed by weakening the rule to the three rows that already comply.

Baseline to preserve or improve: the verifier currently reports 25 PASS and 4 DEBT, the four being permanent negative-route assertions. This feature adds rows; it must not add FAIL or UNDETERMINED.

3. Endpoints added

3.1 GET /api/areas — the taxonomy

Returns published areas. Currently **404** — this endpoint does not exist.

Each area carries: its identifier, title, summary, ordinal, origin (promoted or extracted), evidence certainty (mixed or uncertain-only), the chapters it draws evidence from, counts of its lesson sections and of each question set, and a link to itself.

A3.1.1 — an area with zero evidencing mentions is **never** returned. Publication requires evidence (FR-008), so an unevidenced area appearing here is a contract violation, not an empty edge case.

A3.1.2 — the response reports the count of areas it holds back and why, rather than silently returning a shorter list. A client cannot distinguish “five areas exist” from “eleven exist and six failed publication” unless told.

A3.1.3 — uncertain-only is carried on the wire, not computed by the client. An area resting entirely on uncertain evidence must not be presented as a confident one (FR-012), and leaving that derivation to each client guarantees one of them forgets.

3.2 GET /api/areas/{area} — one area with its materials

Adds to §3.1: the area’s lesson sections in order, each with its identifier, heading, authorship (authored or assembled) and citation count; the publication review record; and links to its question sets, its evidence and its exports.

A3.2.1 — every lesson section states its authorship. Assembled prose is never indistinguishable from authored prose (FR-017).

A3.2.2 — the publication review is present or the area is not served. There is no third condition.

3.3 GET /api/areas/{area}/evidence — where this area appears

Returns the mentions evidencing the area: for each, the passage identifier, the text span, and — for media-backed passages — the time span **and the precision that produced it**.

A3.3.1 — precision is **required** on every entry carrying a time span, and takes exactly word or segment. Omitting it is a contract violation, not a default.

A3.3.2 — where precision is word, the word's timing confidence is carried. It is not discarded because it is usually high: measured, the 5th percentile is 0.559 and 17.4% of words fall below the recorded flag threshold, so “usually high” is precisely the reasoning that produces a confident wrong link.

A3.3.3 — a redacted passage contributes **no** mention and the count of omitted entries is reported, matching how the existing cross-reference endpoint already behaves.

3.4 GET /api/terms and GET /api/terms/{term}

Terms with their canonical form, variants, certainty, owning areas, evidence count, and — on the single-term response — the significance measure **and its inputs**, so the score can be audited rather than trusted (FR-013).

A3.4.1 — an uncertain term is marked on the wire and is never merged into the confident set by the response shape.

A3.4.2 — a term with zero evidence is not returned. If its last evidence was redacted, the term is withdrawn from the taxonomy entirely, and the withdrawal is reflected here rather than only in the display layer.

3.5 GET /api/areas/{area}/questions — the practice sets

Returns the area's short and long sets. Each question carries its identifier, set, kind, prompt, choices where applicable, answer where applicable, category, difficulty, **answer authorship**, the lesson sections it assesses, and **its citations**.

A3.5.1 — the central invariant of this feature. A question is served **only** if it carries at least one citation and **every** citation resolves. A question whose citation resolves to redacted, not present or could not

determine is **withheld** (FR-036, D4). It is not served with a warning, not served with the citation stripped, and not served with an empty citation list.

A3.5.2 — each citation carries enough to jump: the passage identifier, the chapter, the time span where the passage is media-backed, the precision, and a link that lands there.

A3.5.3 — every question in the long set cites **more than one distinct** passage. This is the structural distinction between the sets (D-KG-7); a long-set question with one citation is a contract violation.

A3.5.4 — an assembled answer is marked as assembled wherever it appears (FR-038). A client must not have to infer it.

A3.5.5 — the response reports how many questions were withheld under A3.5.1 and why. A silently shorter set is how a provenance guarantee decays into a claim without anyone noticing.

3.6 GET /api/areas/{area}/coverage — the honest number

Returns the coverage measurement the specification requires: which lesson sections are assessed by at least one question, and the proportion of the area's evidencing passages cited by at least one question.

A3.6.1 — the per-area figures are returned, **never** only an aggregate. An average across areas is how a well-covered area conceals an empty one (SC-017c).

A3.6.2 — **no threshold is expressed** in this response, and no field says "pass". The density of assessable substance per passage has not been measured for this corpus (research U2), so a target here would be a guess that later gets optimised toward.

A3.6.3 — a section with zero questions is listed explicitly, not omitted from the enumeration.

3.7 GET /api/passages/{pid}/knowledge — what this moment teaches

The reverse direction, and the endpoint that makes the recording navigable: given a passage — which a player resolves from a playhead position through the existing transcript window — return the areas, terms, lesson sections and questions attached to it.

A3.7.1 — every entry is reachable in **one** step from this response. The requirement is one-step reachability (FR-019), so a response that returns identifiers a client must resolve separately does not satisfy it.

A3.7.2 — the four outcomes apply to the passage itself. A redacted passage returns the redacted outcome, not an empty knowledge list, because an empty list reads as “*nothing is taught here*”.

A3.7.3 — when nothing is attached, the response says the passage is **unattached and why** (FR-010), rather than returning an empty list. Every passage is accounted for; “no areas” and “not yet classified” are different facts.

3.8 GET /api/graph/traverse — the connectivity matrix, exercised

Traverses from any addressable thing to related things across all four content kinds, in either direction, with a depth bound.

A3.8.1 — traversal is **cycle-safe** and reports truncation rather than stopping silently (FR-026), matching how the existing cross-reference endpoint already reports truncation.

A3.8.2 — no traversal is keyed on a time. Every hop is an identifier hop (FR-033b).

A3.8.3 — a hop whose target cannot be resolved reports the resolution outcome for that hop and continues; it is never dropped from the response, because a dropped hop is indistinguishable from a hop that never existed.

3.9 GET /api/areas/{area}/export — the four formats

Reports which of the four formats exist for an area and links to them.

A3.9.1 — an area without a publication review exports nothing and **says so** (FR-048).

A3.9.2 — when the export toolchain is unavailable, the response is **could not determine**, not an empty format list (FR-049). The measured trap this guards against is real in this repository: a tool that answered a version query and rejected the flag that mattered, because the name on PATH was a symlink into an unrelated cache. Presence of a name proves nothing; capability must be probed.

4. Endpoints changed

4.1 GET /api/search — locus, offsets, new kinds, filters

C4.1.1 — new kinds. area, term and question join the existing kinds. The corpus block that already advertises the indexed kinds must list them, and a kind advertised with zero indexed entries is a **failure** — the corpus currently advertises one such kind, and this feature must not add more.

lesson_section was REQUIRED here and is WITHDRAWN, 2026-09-03, by operator decision. The clause used to list it among the kinds that must be advertised. The implementation deliberately refuses to advertise it, and that refusal is pinned by a test which exists precisely to record it — so the contract and the code knowingly disagreed, and **a contract its own implementation violates on purpose undermines every other clause in the same document.**

The reason it is withdrawn is structural, not “unimplemented”, and that distinction is the whole point. Measured: **0 of 11,622** registry records carry this kind, and its metadata type has **no identifier field** — heading, ordinal, authorship and citation count only. With nothing to key on, a row cannot be minted, so the kind is **not merely unindexed but unkeyable as the data model stands.** Advertising it would breach this clause’s own second sentence, which makes a kind advertised with zero entries a failure.

This is a withdrawal, not a deletion, and it is not a promise that the kind is unwanted. Making it indexable means giving that metadata an identifier — a data-model change rippling through minting, the registry and retrieval, for a kind nothing currently asks for. If that is ever done, restore the requirement here rather than re-deriving it from scratch.

C4.1.2 — locus. Every hit carries a locus: its containing unit and its position within it. A hit whose locus does not resolve is not returned (FR-030).

C4.1.3 — match offsets, the field that is currently a promise. The hit already declares a list of offset pairs and the construction path sets it **unconditionally to empty**; the snippet is generated with empty match delimiters, so nothing anywhere carries an in-string position. Exactly two outcomes are acceptable:

- **populate it** — the lexical leg produces real offsets; or
- **remove it** — and state its absence in the contract.

Leaving a field that promises a position and never carries one is forbidden. It is the blind-instrument shape the governing principles name, and it sits in the precise field the “*link exactly where they appear*” requirement lands on.

C4.1.4 — legs differ, and must say so. The semantic leg does not know which terms matched and cannot produce offsets. It must report that it **cannot** produce them — which is a different fact from producing none. An empty list meaning both is exactly the collapse this contract exists to prevent.

C4.1.5 — filters. Results are filterable by kind and by area (FR-032), and the applied filter is echoed in the response so a client can tell a filtered empty result from an unfiltered one.

C4.1.6 — thresholds survive growth. The existing latency thresholds are not renegotiated by the corpus growing (SC-016, SC-017). The before-and-after figures are published together so a regression is visible rather than merely absent.

4.2 GET /api/suggest — areas and terms

C4.2.1 — area and term names join the suggestion set (FR-031).

C4.2.2 — the suggestion path **still never embeds**. Type-ahead stays lexical; the structural guarantee that the suggestion service holds no embedder is preserved, not weakened for the new kinds. The measured basis is unchanged: semantic embedding under load has been observed in the tens of seconds, which cannot meet a 200 ms budget.

4.3 POST /api/ask — bounded by an open clarification

C4.3.1 — retrieval over the new kinds is extended. This is safe because the retrieval gate refuses before any model runs.

C4.3.2 — generation about areas, lessons and questions is gated on the third clarification. Until it is answered, this endpoint does not generate answers about the new kinds.

C4.3.3 — if generation is enabled, a verification layer checking the answer against the **question** must exist and be measured first (FR-051). The existing four layers all verify the claim against the **passage**; none verifies it against the question, which is why a topically related fragment that does not answer what was asked passes all four.

C4.3.4 — a verifier that was requested and could not load reports **could not determine**. It does not degrade silently to the weaker verifier (FR-052).

4.4 GET|POST /api/progress — areas and question sets

C4.4.1 — progress covers areas and question sets in addition to what it already covers.

C4.4.2 — progress **survives a content re-ingest** (FR-039). The reference module deliberately preserves its progress table while wiping everything else on rebuild; that behaviour is ported. A learner's history is not a derived artifact.

5. Gates and their paired mutations

Every gate here owes a proof that it fails when its condition is broken (FR-053, SC-027). A gate never observed failing is not known to work.

Gate	Asserts	Paired mutation that must turn it red
G-KG-1	every endpoint in §3 has a manifest row and answers per R1–R3	remove one manifest row
G-KG-1-changed	every endpoint in §4 has a manifest row whose contract citation names both its contracts/http-api.md section and its	rewrite one changed endpoint's contract citation back to its 001-only form, leaving the row otherwise byte-identical

Gate	Asserts	Paired mutation that must turn it red
	§4 subsection here (R1b)	
G-KG-2	a question with an unresolvable citation is withheld	serve it with the citation stripped
G-KG-3	every long-set question cites more than one distinct passage	admit a single-citation question to the long set
G-KG-4	every time-carrying entry declares its precision	omit precision and default it to word
G-KG-5	offsets are populated, or the field is absent and its absence stated	return an empty offset list from a leg that could produce offsets
G-KG-6	no relationship is keyed on a time	introduce one time-keyed link
G-KG-7	redaction reaches all eight propagation targets	skip exactly one target
G-KG-8	an unresolvable link fails loudly	make the resolver re-point by nearest match
G-KG-9	an unavailable export toolchain yields could-not-determine	return an empty format list instead
G-KG-10	an unevidenced area or term is never served	publish one with its mentions removed
G-KG-11	a kind advertised as indexed has retrievable content	advertise a kind with an empty index
G-KG-12	coverage is reported per area, never only as an aggregate	replace the per-area table with a mean

G-KG-1-changed is a separate gate from G-KG-1, and its paired mutation is a rewrite rather than a deletion, because a deletion cannot reach the defect. G-KG-1's mutation removes a manifest row and the gate goes red — correct for §3, where the failure is an endpoint with no row at all. It proves nothing about §4, where **the row is present throughout**. Every §4 route already had a row, carried over from specs/001, and the regression R1b guards against is one field inside a row that never went missing: the contract citation silently reverting to its 001-only form while the route keeps its locus field, its new kinds, its filters and its new refusal vocabulary. A gate whose only mutation is row deletion is a blind instrument for this class, and it would report PASS. The mutation must therefore be the regression **as it would actually occur** — a surgical edit of the contract column, with the rest of the row untouched — and the gate must go red on it and green again on restore.

G-KG-11 is stated as retrievability, not as a row count, on purpose. A row count cannot distinguish a populated index from a populated table nobody queries, and the corpus already contains one advertised kind with zero entries — which a row-count check would have caught only by accident and a retrieval check catches by construction.

6. Traceability

Requirement	Where in this contract
FR-019, FR-020	§3.3, §3.7
FR-021	§3.3 A3.3.1–2, §5 G-KG-4
FR-023, FR-024	§1.2, §3.8 A3.8.3, §5 G-KG-8
FR-026	§3.8 A3.8.1
FR-027	§5 G-KG-7
FR-028 to FR-033	§4.1, §4.2
FR-033a, FR-033b, FR-033c	§3.8, §4.1
FR-034a, FR-034b	§3.5 A3.5.3, §3.6
FR-035, FR-036	§3.5 A3.5.1, §5 G-KG-2
FR-037	§3.5 A3.5.2
FR-038	§3.2 A3.2.1, §3.5 A3.5.4
FR-039	§4.4 C4.4.2
FR-048, FR-049	§3.9
FR-050 to FR-052	§4.3

Requirement	Where in this contract
FR-059	§2

Requirements deliberately not covered here, and where they live: taxonomy extraction and promotion (FR-006 to FR-013), materials authorship and review (FR-014 to FR-017), the repeatable pipeline (FR-033d to FR-033g), the learning surface and its visual identity (FR-040 to FR-044b), and the governance requirements (FR-053 to FR-058) belong to the knowledge-graph contract, the pipeline, the front end and the platform’s existing governance instruments respectively. They are listed so their absence here reads as a boundary rather than as an omission.

7. Unverified register

- **U1 (from research)** — whether the full-text engine exposes match positions through a supported interface, or whether offsets must be recomputed by re-locating query terms in the matched text. This decides whether C4.1.3 is cheap or a second pass, and it has a correctness edge: a recomputation can disagree with what the index actually matched, particularly under the prefix matching the lexical leg applies to the final query token. **C4.1.3 must not be implemented on an assumption about this.**
- **U4 (from research)** — whether an entailment model loads on this host. C4.3.3 and C4.3.4 depend on it. The platform’s code claims it refuses to degrade silently; that claim has not been observed for this feature, and a claim is not a measurement.